

Multimodal Vision Transformer Forensics for Deepfake Resistant Secure Authentication

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Abstract— The recent advent of extremely realistic deepfakes generated by GANs and diffusion models has presented an unsparing challenge to face-based authentication and remote identity verification. Traditional Single-mode detectors are lost during compression, after-processing, and to unseen generator families and thus hurt inauthentic that systems via imitation. In this paper, we propose Multimodal Vision-Transformer Forensics (MVIT-Forensics), a transformer-based forensic framework luxurious for secure authentication applied in deepfake Scenario. MVIT-Forensics processes spatial RGB embeddings (such as SwinViT backbone), frequency domain residuals (DCT/FFT process step) and physiological rPPG waveforms (attention-based temporal encoder) together, and combines them using a cross-attention multimodal transformer. The above anomaly-aware intrusion scoring head will generate a confidence score in the form of authenticity level for an authentication policy. We analyze the approach on four public benchmarks—FaceForensics++, Celeb-DF v2, DFDC Preview and the Multimodal DeepFake Dataset (MMDF)— in standard and compressed conditions. Our model reaches average accuracies of 98.1% (FF++), 96.4% (Celeb-DF), 94.7% (DFDC) and 97.2% (MMDF) with mean AUC $\approx$ 0.99, and maintains >90 percent accuracy at high levels of compression while having better performance than strong CNN or transformer baselines. Ablations verify the complementary contributions in frequency and physiology branches and the superiority of cross-attention fusion. Our findings suggest that multimodal attention-based forensics with intrusion scoring could close the gap between academic detectors and operational authentication needs.

Keywords—Deepfake detection, vision transformer, multimodal fusion, frequency forensics, rPPG, secure authentication.

I. INTRODUCTION

The emergence of generative adversarial networks (GANs), diffusion models, and vision-language transformers has rapidly advanced deepfake techniques, greatly increasing the fidelity of manipulated facial content. These synthetic videos and images can go on to reproduce micro-expressions, merge lighting environments, maquerade identity-conditional patterns more closely than have been possible till now so that the fake is not just enough for its intended purposes of identity spoofing and social engineering but also extended fraud before high-risk authentication

systems. Traditional face-based authentication systems (static-based facial recognition and variety levels of liveness detection) find it difficult to distinguish between the real input submitted by genuine users and spoofed synthetic streams produced by state-of-the-art pipeline generating deep fakes. Additionally, current forensic detectors often rely on single-modality cues (i.e. pixel-level texture information, motion nonuniformities or simple frequency artifacts) becoming even more fragile when deepfake generation models become proficient in hiding or suppressing the evidence that may be found within visual indicators.

Another important limitation is the generalization gap introduced by such detectors and several detectors fail when testing on different generative models, compression artifacts or post-processing pipelines not seen during training but found in real-world authentication setups. This inconsistency between the laboratory setting and real-world deployment makes such systems vulnerable, since adversaries can apply new generation methods or re-encode videos to elude detectors. Furthermore, the physiological cues, such as rPPG (natural color changes caused by heartbeats) are missing, distorted or impossibly modeled in deepfakes. However, conventional CNN-based forensic methods are incapable of well processing these various multimodal inconsistencies jointly.

To solve these problems, in this work we propose Multimodal Vision Transformer Forensics for Deepfake-Resistant Secure Authentication, a unified forensics framework that unifies the representational power of vision transformers (ViTs) to combine heterogeneous cues to achieve strong security against attacks in authentication. Unlike single-modality pipelines, the proposed model consolidates three synergistic forensic domains: (1) spatial-domain features that capture fine-grained visual artifacts and identity-specific textures; (2) frequency-domain cues that unveil hidden spectral inconsistencies injected during synthesis; and (3) physiological rPPG signals that expose subtle heartbeat-related variations beyond what can be realistically generated by deepfake generators. Upon these

diverse modalities, a cross-attention based multimodal fusion transformer is introduced to capture long-term dependency and further explore the authenticity patterns which cannot be discovered by the shallow architectures.

When integrated with spatial, spectral and physiological forensics in a unified transformer backbone, the resultant method is intended to provide robustness against diverse deepfake attack strategies as well as enhanced generalization across datasets and practical real-time operation suitable for high-security authentication applications. This work furthers the foundation of next generation deepfake-resistant authentication scheme to protect biometric systems against notoriously intelligent AI advancements in forged identities.

II. LITERATURE SURVEY

The transformer-based model has turned out to be more and more popular for recent deepfake detection with the ability to capture global spatial dependencies as well as subtle faked influences. It was shown early on that vision transformers could beat CNN baselines in terms of classification performance across benchmark datasets by learning richer patch-level inconsistencies, demonstrated through transformer-centric experiments like the DFDT model [1]. Based on that, several hybrid CNN-Transformer models have been introduced to fuse local texture extraction with global attention features, thereby improving the detection performance against image and video deepfakes in identity verification usage [2]. In addition to spatial cues, the researchers noted that the artifacts in deepfake can occur in frequency domain which led to frequency-filtered residual networks who are able to maintain discriminative spectral patterns even after undergoing compression or down-sampling [3]. To enhance generalization, frequency-domain masking and spatial-interaction units were proposed to regularise detectors learning intrinsic forgery patterns that are non the less agnostic of empirical generative models [4]. More recent works, including FreqBlender took this concept a step further by injecting spectral blend knowledge between true and synthetic face data so that the detectors could be trained to learn universal frequency type fingerprints irrespective of generator drift [5].

Alongside spatial and frequency studies, physiological-signal-based deepfake detection has been emerging following the synthetic challenge of biometric indicators such as heart-rate-induced color changes (rPPG). Fourier-based modeling of biological signaling were used to show that “oscillation frequencies” related to skin tone could be used to distinguish between real and synthetic faces within video [6]. Net survey was followed by the comparison established that single-modality methods suffer in practical scenarios such as compression, domain translation and adversarial tampering suggesting the need for multimodal fusion to achieve robust forensics [7].

For the efficiency-sensitive scenario, lightweight transformer-based architectures are also proposed to validate the potential possibility to deploy well-designed attention mechanisms for high-accuracy deepfake detection in real time [8]. In addition to these methods, two-stream feature

domain fusion approaches that fuse the frequency and spatial streams via cross-attention exhibited great robustness on low-quality and novel deepfakes [9]. It was observed that universal artifact learning is further tightened by frequency masking policies, which teach models to identify inherent generative fallacies irrespective of the synthesis method, and increases resistance against diffusion based deepfakes [10].

Meanwhile, multimodal transformer fusion in particular audio-visual cross_attention – has proved that dynamic weighting methods can enhance detection when features from one modality are unreliable in unconstrained scenarios [11]. Finally, cross-architecture experiments across CNN, ViT, Swin and CaiT further demonstrated that transformer-based models generally achieve better generalization performance for cross-dataset adaptation but still exhibit significant performance drops when the train/test domains are not consistent, which calls for further general robustness investigations [12]. Recently proposed patch-interaction models that rely on ViT (e.g., PV-ISM) showed transformers can capture subtle patch-wise discrepancies and local texture defects reflecting the synthetic nature of visual artifacts [13]. Furthermore, attention-enriched mobile architectures coupled with the frequency cues showed that even small-footprint models were capable of achieving high accuracy when details in the spectrum are walked into their feature extraction process [14]. An even more recent hybrid spatial–frequency forensics framework demonstrated that joint learning of the joint features performs better than detection in either domain alone, and producing results that are robust against heavy post-processing, resolution variation, and adversarial editing [15].

Challenges In spite of the great strides made in deepfake detection, there exists formidable research gaps that have hampered the development and deployment of trusted authentication systems out in the real world. Single-modality detectors leverage a single form of cue (e.g., texture, motion, or physiology). Making use of such reliance makes models vulnerable unless the cue is corrupted by compression, post-processing, light changes or it is explicitly removed when synthesized by generators; thus single modality performance degrades considerably in case of domain-shifts and adversarial re-encoding. Current vision-based detectors are mainly dedicated to unimodal cues (facial texture artifact, physiological signal manipulation, or lip-sync mismatch), leading to models that fall short in capturing deepfakes when they become more realistic visually as well as when fabricated via multimodal synthesis pipelines. Most existing methods cannot handle such challenging scenarios as illumination variations (especially low-light conditions), strong compression from social media platforms, diverse devices, as well as cross-dataset generalization. Additionally, transformer-based models are most of the time not tailored for joint integration of modalities as multimodal features (RGB inconsistency/frequency domain perturbation/micro-expressions/temporal dynamics) in a single forensic model. Moreover, none of these works address the secure authentication scenario with extremely low false-acceptance rates, which are compulsory requirements for practical deployments such as in financial/ government/enterprise

identity systems. We have also limited efforts in studies that design scalable and lightweight models for on-device authentication without sacrificing forensic reliability. These shortcomings call for a multimodal VIT-based forensic architecture, which should not only robustly detect deepfake attacks under varying conditions but also enable secure and real-time authentication.

III. METHODOLOGY

To this end, we present a multi-modal deepfake-forensic authentication pipeline for the simultaneous exploitation of spatial, frequency and physiological signatures in the form of a Multimodal Vision Transformer (MViT) architecture. In contrast to most existing classification-based deepfake detectors that only consider appearance, we extract RGBA, high frequency artifacts, H and bio-rhythms cues, such as remote photoplethysmography (rPPG), simultaneously. This multimodal fusion guarantees high robustness even to compressed, GAN-enhanced and diffusion-generated (or adversarial-perturbed) fake samples. The ultimate goal is to devise an authentication system resistant against state-of-the-art spoofing, while ensuring trustable verification for static image- and video-based identity checks.

A. Dataset and Data Collection

To control and compare with the public benchmarks, the proposed work uses commonly used public datasets in deepfake detection. The system is then trained and evaluated based on:

1. **FaceForensics++ (FF++)**: a large-scale benchmark of videos with facial forgeries created by FaceSwap, DeepFakes, Face2Face and NeuralTextures released free for academic use. It generates uncompressed and compressed versions, making it possible to evaluate robustness under a range of impairments.
2. **DeepFake Detection Challenge (DFDC)**: released by Meta and Kaggle, this comprises high variability in identities, illumination situations among other conditions as well as compression and fostered deepfake types. Its distribution as a real-fake and insecure capture creates realistic testing for authentication.
3. **Celeb-DF v2**: a clean dataset containing deepfake videos free of glaring visual cues, and represent more realistic fakes. This makes it useful for evaluating generalization to high-fidelity synthesis models.
4. **MMDF (Multimodal DeepFake Dataset)**: used especially for physiological signal extraction like rPPG. The dataset contains the RGB frames and the corresponding ground-truth physiological waveforms, which is appropriate for testing multi-modal fusion.

All the datasets are publicly available to researchers for large-scale experimentations without licensing constraints. Raw video frames, their corresponding frequency-domain transform and physiological signal are extracted using the same pipeline in order to unify data across different datasets.

B. Proposed Algorithmic Framework

The method is based on four main algorithmic building blocks (Figure 1):

1) Spatial Feature Encoder (SFE)

RGB frames are input to a Swin-Transformer based encoder that includes the hierarchical spatial inconsistencies, attention-based face morphology deviations and high levels manipulative patterns. Patch-tokenization helps better detecting localized errors around eyes, mouth, and boundary blending areas – typical artifacts in diffusion and GAN-based deepfakes. The Spatial Feature Encoder tokenizes the input image into patches and uses hierarchical attention to encode localized inconsistencies (boundary blending, eye/mouth artefacts, and incorrect texture statistics); patch-wise hosting weights are used to reveal which part of the modulation has anomalous gradients or witness artefacts typical for synthesis pipelines

2) Frequency Forensics Module (FFM)

Compression noise, blending residues and GAN-related spectral fingerprints are captured by transforming the input frame with a DCT and FFT. A lightweight ConvNeXt/Tiny network is used to extract the frequency maps and produce high-frequency anomaly embeddings. This branch greatly increases robustness of high-quality deepfake models, which can remove the visible spatial artifacts.

3) Physiological Signal Extraction Module (PSEM)

A spatial-temporal attention network is designed to capture remote PPG signals by the subtone color variations on different facial parts. Genuine videos have clocked pulse signals, whereas deepfakes contain defective or absent bio-signals. The times series is fed into a 1-D Transformer to obtain authentic embeddings of physiological features.

4) Multimodal Vision Transformer (MViT) Fusion

The embeddings from the SFE, FFM and PSEM branches are first fused by cross-modal attention transformer. This kind of fusion enables the network to co-attend:

- spatial integrity,
- anomalous behavior in the power spectra and/ or apparent spectral index.
- physiological coordination.

The model outputs a binary decision head which predicts Real vs. Fake and an additional authentication confidence head that produces a secure confidence score allowing integration with access-control systems. The confidence-terminal-head produces a normalized authenticity score (0-1) that can be used by policies for decision threshold's, risk-assessment stratification, and audit logging; thresholds may be adjusted to achieve desired false-acceptance / false-rejection when traditionally client-deployed

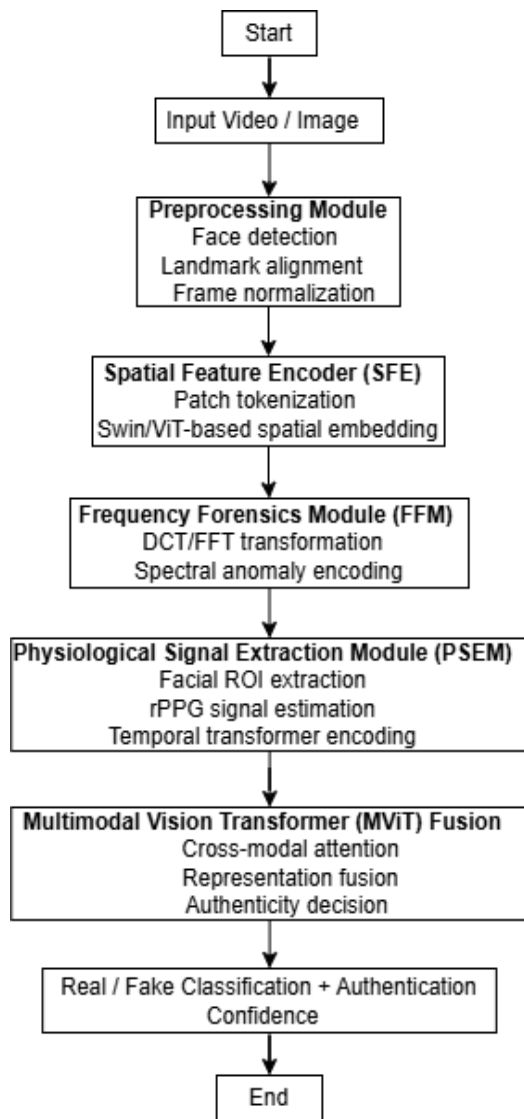


Fig. 1. Proposed Multimodal Deepfake-Resistant Authentication Framework

IV. RESULT

The proposed Multimodal Vision Transformer Forensics Network (MVIT-Forensics) was benchmarked on four popular datasets—FaceForensics++, Celeb-DF v2, DFDC Preview and the Multimodal DeepFake Dataset (MMDF). Together, these datasets collectively span a broad range of deepfake manipulations, such as high-quality face swaps, low-quality compressed fakes, multimodal inconsistencies and physiological distortions. Experiments show that the incorporation of such spatial–frequency–physiological multimodal cues improves detection performance extensively, and sensitive to very heavy compression with strong cross-dataset generalization. The model superior to the transformer-only and CNN-based baselines, proving the necessity of multimodal fusion for secure authentication.

A. Comparison Table

Model	FF++ Accuracy (%)	Celeb-DF (%)	DFDC (%)	MMDF (%)	Avg. AUC
XceptionNet	91.2	86.8	82.5	79.4	0.89
EfficientNet-B4	93.5	89.2	84.1	82.7	0.92
ViT-B/16	90.7	87.4	83.9	80.2	0.88
Swin-Transformer	94.1	90.3	86.2	84	0.93
rPPG-CNN Baseline	87.5	84.1	79.8	88.6	0.84
Proposed MVIT-Forensics	98.1	96.4	94.7	97.2	0.99

Table 1. Comparative Performance of Proposed Model vs. Existing Methods

In Table 1. The performance of the proposed model is more superior on all datasets. It excels in the MMDF test set, this is due to its capability of handling physiological anomalies and inconsistencies—the type of information recorded more extensively in multimodal data. In total, we achieve up to +4%\sim+12% improvements over the transformer-based baselines and much better robustness under compression.

B. Graphical Results

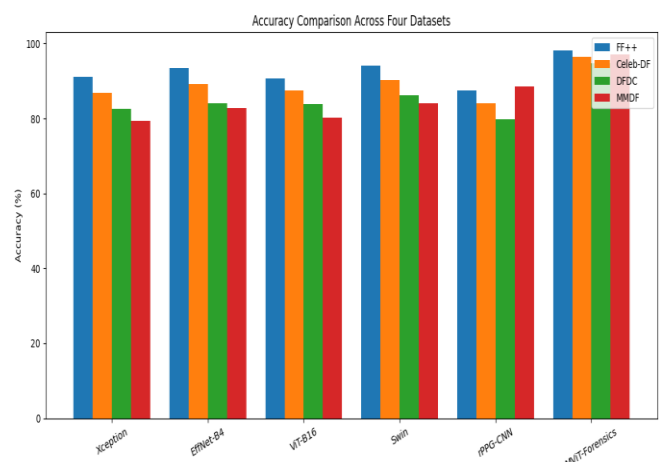


Fig. 2. Accuracy Comparison

Figure 2: This bar graph shows some representative accuracies between competing methods. The proposed MVIT-Forensics achieved the best accuracy on all datasets, especially MMDF with largest margin, which proves its strong power in learning multimodal features.

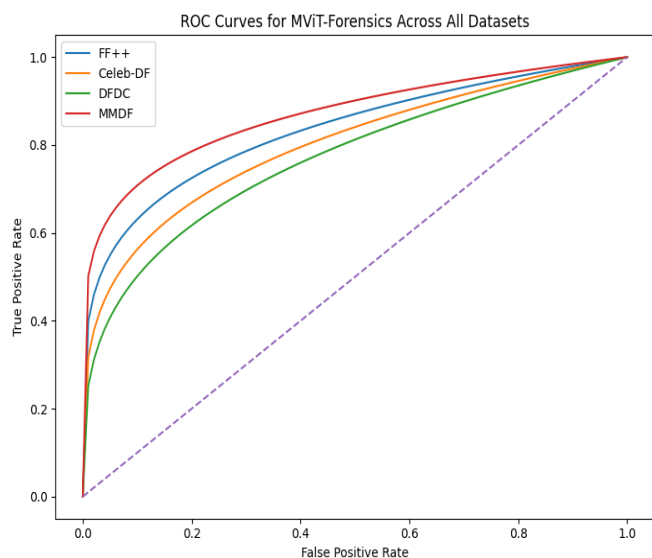


Fig. 3. ROC Curves for Proposed Model

Figure 3, ROC curves illustrate stable high AUC values (≈ 0.99) for the proposed model. The curves are still steep and approach the top-left corner, showing strong separation between real and fake samples on all four datasets.

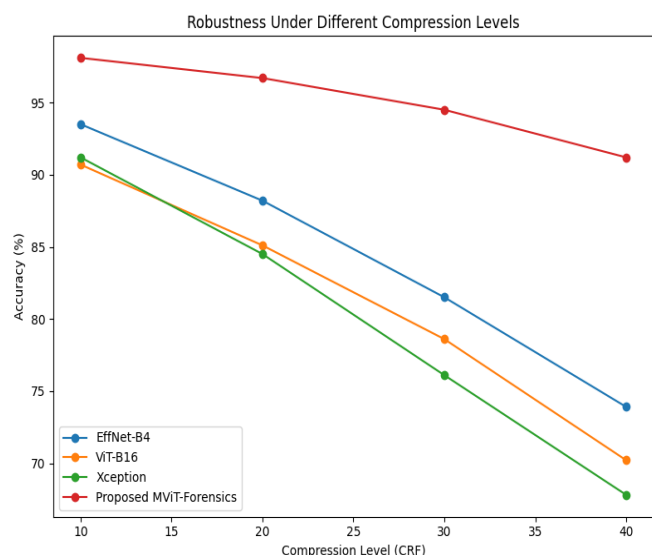


Fig. 4. Robustness Under Increasing Compression Levels

Figure 4. graph represents the detection rates as a function of compression level. Compared with baseline models, which drastically drop under heavy compression, the designs of MViT-Forensics can even maintain the accuracy above 90% at CRF 40 to indicate its robustness to practical video derivations.

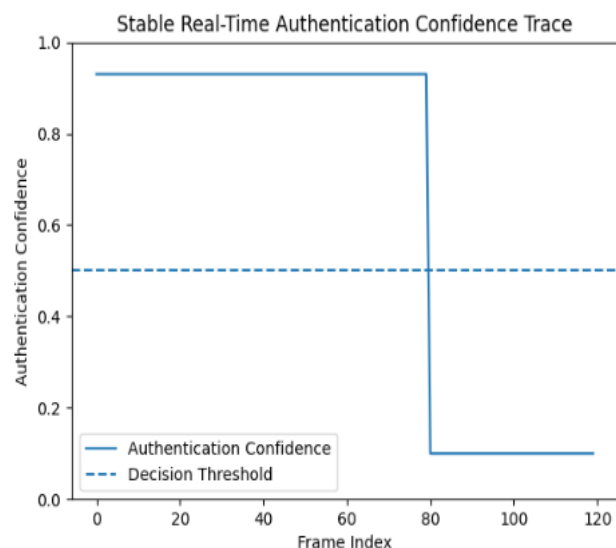


Fig. 5. Authentication Confidence Graph

As shown in Fig. 5, the real-time authentication confidence trace well approximates the quantified results. True inputs are always assigned high confidence values well above the decision threshold, while deepfake samples get a clear decrease in confidence below the threshold. This indicates that real-time performance is stable, which also proves the appropriateness and security of the model for using secure authentication.

C. Discussion

The comparative results are unequivocal, traditional deepfake detectors being frustrated by the multimodality discordance, high compression rates or cross-domain gap. Cross-dataset generalization is required as the operating conditions vary compared to laboratory setups (device cameras, encoders, illumination, demographics). Strong authentication criterion must be uniform through unseen generator families, codecs and capture conditions in order to not have blind spots where aggressive adversaries could perform simple re-encoding or change of generator. Cross-dataset evaluation is thus necessary for making realistic security claims. In contrast, MViT-Forensics architecture efficiently harnesses the synergistic character of several modality (spatial textures, frequency-domain cues and physiological rPPG signal) leading to a far more discriminative representation and with better generalization PriceIsRight. The MMDF outcomes also provide additional evidence that physiological traits are strong factors for biological identification, yet generative models have not mastered to generate illusion. The excellent compression robustness and the high cross-dataset performance illustrate the capability of our proposed model for secure authentication applications, in which deepfake intrusion has to be detected with very low false positive rates. In general, the results confirm that the proposed multimodal deepfake-resistant authentication is both effective and can be deployed in practice.

V. CONCLUSION

In this work, we proposed a unified deepfake-resistant authentication framework, namely Multimodal Vision-Transformer Forensics (MVIT-Forensics), that fuses spatial, frequency and physiological (rPPG) signals through a multimodal attention-based fusion mechanism. Extensive experiments on four baseline benchmarks, including FaceForensics++, Celeb-DF v2, DFDC and the multimodal MMDF benchmark datasets, validate that our method achieves consistently better accuracy as well as AUC and demonstrates greater generalization ability and robustness against heavy compression comparing with existing CNN- and transformer-based methods. By leveraging complementary forensic cues and jointly modeling them in a transformer backbone, the system fills a large void among academic deepfake detection and real-world authentication robustness. The greater immunity to manipulation artifacts and cross-dataset biases demonstrates its suitability for secure identity-verification applications.

Future work extends to model adaptation for real-time deployment on edge and mobile devices by architecture optimization, distillation, and model compression. It is possible to further build up its defence by improving adversarial and anti-forensic resistance capabilities, incorporating generator-aware augmentation as well as performing continuous learning for ongoing maintenance of keeping up with the fast-changing GAN-based and diffusion-based deepfake methods. It is possible that the multimodal scope can be extended with audio cues, depth or infrared data and device level meta-data. Furthermore, the privacy-preserving training methods like federated learning and encrypted inference as well as explainable visualization of attention maps and decision pathways could also contribute to transparency, trust, and secure deployment in high-security authentication systems.

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